# A Hybrid Intrusion Detection System using Explainable AI for Enhanced Accuracy and Transparency

**Publisher: IEEE** Cite This PDF

A. Hemalatha ; Vimal Kumar M N ; F. Twinkle Graf ; Anlin Sahaya Infant Tinu M ; P. Pavithra ; R. Suresh **All Authors**



**Abstract:**
Intrusion detection system is highly effective and easy to understand in the situation of ubiquitous cyber threats. The trustworthiness and interpretability of traditional intrusion detection systems are balanced due to their dependence on transparent deep learning (DL) and Machine Learning (ML)methods. To overcome these issues, this research proposes a hybrid IDS which makes use of Explainable AI(XAI). An integration of ML methods like random forest and support vector machine with DL methods like CNN ensure efficient detection accuracy in the system. The decision-making process can be better efficient with the help of XAI methods like SHAP and LIME, which increases transparency. The CICIDS 2017 dataset is used to validate the feature extraction of framework, model training and interpretability components. The result show that proposed method performs better with 98.5% accuracy rate and better precision, recall and F1-socre when compared to existing methods. This method provides useful data for securing networks while reducing the number of false positives and negatives. Through the integration of advanced machine learning/deep learning methods with XAI, the proposed IDS ensures dependability, openness and practicality.

**Published in:** 2025 International Conference on Electronics and Renewable Systems (ICEARS)

**Date of Conference:** 11-13 February 2025

**DOI:** 10.1109/ICEARS64219.2025.10940840

**Date Added to IEEE *Xplore*:** 02 April 2025

**Publisher:** IEEE

**ISBN Information:**

**Conference Location:** Tuticorin, India



## Authors

A. Hemalatha
Department of Information Technology, E.G.S. Pillay Engineering College, Nagapattinam

Vimal Kumar M N
Department of Mechatronics Engineering, Sona College of Technology, Salem

F. Twinkle Graf
Department of Computer Science & Engineering (Data Science), Madanapalle Institute of Technology & Science, Madanapalle

Anlin Sahaya Infant Tinu M
Department of Biomedical Engineering, Rohini College of Engineering and Technology, Anjugramam

P. Pavithra
Department of Computer Science and Engineering, TJS Engineering College, Kavaraipettai, Peruvoyal, Gummidipoondi

R. Suresh
Department of Electronics and Communication Engineering, TJS Engineering College, Kavaraipettai, Peruvoyal, Gummidipoondi

Figures

References

Keywords

Back to Results